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Power supply system for detecting a rail violation in a multi-rail sequence

Abstract

In some aspects, a power supply system includes a plurality of power supply devices configured to be connected to an integrated circuit, where a power supply device of the plurality of power supply devices includes a voltage regulator configured to generate a rail voltage, and an internal diagnostic circuit. The internal diagnostic circuit is configured to detect activation of an enable signal to activate the plurality of power supply devices according to an activation power sequence, detect a rail violation of the activation power sequence in response to a value of the rail voltage at a first time, not satisfying a first condition defined by a voltage threshold, or at a second time, not satisfying a second condition defined by the voltage threshold. In response to the rail violation being detected, the internal diagnostic circuit is configured to activate an interrupt signal.

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Background/Summary

BACKGROUND

(1) An integrated circuit (e.g., a system-on-chip) may use multiple voltage supply rails that supply power to the integrated circuit. The voltage supply rails may power up and power down in a pre-defined sequence. Deviating from the predefined sequence may impact reliability and functional safety of the integrated circuit.

SUMMARY

(2) In some aspects, the techniques described herein relate to a power supply system, including: a plurality of power supply devices configured to be connected to an integrated circuit, a power supply device of the plurality of power supply devices including: a voltage regulator configured to generate a rail voltage; and an internal diagnostic circuit configured to: detect activation of an enable signal to activate the plurality of power supply devices according to an activation power sequence; detect a rail violation of the activation power sequence in response to a value of the rail voltage: at a first time, not satisfying a first condition defined by a voltage threshold, or at a second time, not satisfying a second condition defined by the voltage threshold; and in response to the rail violation being detected, activate an interrupt signal.

(3) In some aspects, the techniques described herein relate to a power supply device including: a voltage regulator configured to generate a rail voltage; and an internal diagnostic circuit configured to: detect activation of an enable signal to activate a plurality of power supply devices according to an activation power sequence; detect a rail violation of the activation power sequence in response to a value of the rail voltage: at a first time, not satisfying a first condition defined by a voltage threshold, or at a second time, not satisfying a second condition defined by the voltage threshold; and in response to the rail violation, activate an interrupt signal.

(4) In some aspects, the techniques described herein relate to a method of monitoring a power sequence of a power supply system, the method including: in response to activation of an enable signal to activate a power supply system according to an activation power sequence, activating a power supply device to generate a rail voltage; detecting a rail violation of the activation power sequence in response to a value of the rail voltage: at a first time, not satisfying a first condition defined by a voltage threshold, or at a second time, not satisfying a second condition defined by the voltage threshold; and in response to the rail violation being detected, activating an interrupt signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1**A illustrates a power supply system for detecting a rail violation of a power sequence of a plurality of power supply devices.

(2) FIG. **1**B illustrates an example of the power supply system.

(3) FIG. **1**C illustrates an example of a power supply device with an internal diagnostic circuit.

(4) FIG. **1**D illustrates an example of the internal diagnostic circuit.

(5) FIG. **1**E illustrates a diagram depicting example signals for detecting a rail violation by a power supply device.

(6) FIG. **1**F illustrates a diagram depicting example signals for detecting a rail violation by a plurality of power supply devices.

(7) FIG. **2** illustrates an example of a power supply device for detecting a rail violation.

(8) FIG. **3** illustrates another example of a power supply device for detecting a rail violation.

(9) FIG. **4** illustrates a flowchart depicting example operations of a power supply device for detecting a rail violation.

DETAILED DESCRIPTION

(10) This description relates to a power supply system configured to detect, by an individual power supply device, a rail violation of a power sequence of a plurality of power supply devices connected to an integrated circuit (e.g., a system-on-chip (SoC)). For example, the power supply system may include a plurality of power supply devices (e.g., point of load (POL) regulators), where each power supply device is configured to generate a separate rail voltage that is used by the integrated circuit. In response to an enable signal, the power supply devices may activate (e.g., power up) and deactivate (e.g., power down) in a power sequence (e.g., pre-defined sequence) that

defines the timing of when a power supply device is activated. For example, the power sequence may include activating a first rail voltage, which is followed by a second rail voltage, and then followed by a third rail voltage. Deviating from the power sequence may be referred to as a rail violation (e.g., the second rail voltage powers up before the first rail voltage reaches its target value). In some systems, rail violations may impact reliability and/or functional safety of the integrated circuit.

(11) A power supply device may include a sequence delay counter and a voltage regulator configured to generate the rail voltage. The sequence delay counter may trigger the voltage regulator to activate or deactivate according to programmable counter values such as a power-up delay value and/or a power-down delay value. The sequence delay counter may receive the enable signal, and, in response to activation of the enable signal, the sequence delay counter may trigger the voltage regulator to activate (e.g., power-up) when the time (from receipt of the activation of the enable signal) corresponds to the power-up delay value, which causes the rail voltage to increase to its target value. In response to deactivation of the enable signal, the sequence delay counter may trigger the voltage regulator to deactivate (e.g., power-down) when the time (from receipt of the deactivation of the enable signal) corresponds to the power-down delay value, which causes the rail voltage to decrease.

(12) The power system discussed herein uses the power supply devices themselves to monitor, control, and/or diagnose the power sequence, thereby providing a more flexible, easier-to-implement power system that may avoid using a central management unit (e.g., power management integrated circuit (PMIC)) to monitor and control the power sequence. In some examples, a central management unit may be customized to a certain integrated circuit, and, if the central management unit controls and monitors the power sequence of the power system, implementing a change to the power system (e.g., replacing a power supply device, etc.) may be relatively difficult.

(13) Instead of using a central management unit, in some implementations, each power supply device includes an internal diagnostic circuit configured to detect a rail violation of its respective rail voltage (e.g., during activation and deactivation of the power system, and, in some examples, when the rail voltage is active), and, when detected, the internal diagnostic circuit may generate an interrupt signal (e.g., safe state signal), which may cause the integrated circuit to operate in a safe state. For example, the internal diagnostic circuit may detect whether its respective rail voltage has started up too early or too late in the power sequence by monitoring the rail voltage during a specified monitoring window and/or at specific times.

(14) In response to activation of the enable signal, the internal diagnostic circuit may detect whether a value of the rail voltage satisfies a first condition defined by a first voltage threshold (e.g., an undervoltage threshold) at a first time (e.g., the start of the monitoring window) to ensure that the power supply device has not started-up too early. In some examples, the value of the rail voltage being equal to or greater than the first voltage threshold may indicate a rail violation (e.g., started power up too early). Also, during activation, the internal diagnostic circuit may detect whether the value of the rail voltage satisfies a second condition defined by the first voltage threshold at a second time (e.g., the end of the monitoring window) to ensure that the power supply device has not started up too late. The value of the rail voltage being equal to or less than the first voltage threshold at the second time may indicate a rail violation (e.g., powered up too late).

(15) In response to deactivation of the enable signal, the internal diagnostic circuit may detect whether the value of the rail voltage satisfies a first condition defined by a second voltage threshold (e.g., a disabled threshold) at a first time (e.g., the start of the monitoring window) to ensure that the power supply device has not powered down too early. At the first time, the value of the rail voltage being equal to or less than the second voltage threshold may indicate a rail violation (e.g., powered down too early). During deactivation, the internal diagnostic circuit may detect whether the value of the rail voltage satisfies a second condition defined by the second voltage threshold at a second time (e.g., the end of the monitoring window) to ensure that the power supply device has

not powered down too late. At the second time, the value of the rail voltage being equal to or greater than the second voltage threshold may indicate a rail violation (e.g., powered down too late). These and other features are further described with reference to the figures.
(16) FIGS. **1**A through **1**F illustrate a power supply system **100** configured to provide rail voltages **106** to an integrated circuit **104**. In some examples, the integrated circuit **104** includes a system-on-chip (SoC). At least one (e.g., each) power supply device **102** may monitor its respective rail voltage **106** to detect a rail violation **134** to the power sequence of the power supply system **100**. In response to the detection of a rail violation **134**, a power supply device **102** may activate an interrupt signal **110**, which causes the integrated circuit **104** to operate in a safe state.
(17) A power sequence is the timing order in which the power supply devices **102** are activated (e.g., powered up) or deactivated (e.g., powered down). A rail violation **134** may indicate an abnormality to the power sequence (e.g., a particular rail voltage **106** being powered up (or down) too early or too late in the power sequence). Each of the power supply devices **102** may be configured to detect whether its respective rail voltage **106** is being powered up (or powered down) at a time that is too early or too late in the power sequence. The power supply system **100** may not include or use a central management unit (e.g., a power management integrated circuit (PMIC)) to monitor and control the power sequence, which may provide more flexible and easier-to-implement power supply devices **102**.
(18) The power supply system **100** includes a plurality of power supply devices **102** configured to generate a plurality of rail voltages **106**. The rail voltages **106** may be the supply voltages used by the integrated circuit **104**. In some examples, each power supply device **102** has a voltage output that is connected to the integrated circuit **104**. A power supply device **102** may be any type of electrical device that generates a rail voltage **106** from a supply voltage **116**. In some examples, the power supply devices **102** share a supply voltage **116** (e.g., a common supply voltage). Each power supply device **102** may receive the supply voltage **116**, and, when activated, may generate separate (and, in some examples, different) rail voltages **106**. During activation of a power supply device **102**, a rail voltage **106** increases from a minimum voltage (e.g., zero volts, standby voltage, etc.) to a target voltage. During deactivation of one of the power supply devices **102**, a rail voltage decreases from its target voltage to its minimum voltage. In some examples, the power supply devices **102** include voltage regulators (e.g., voltage converters). In some examples, the power supply devices **102** include direct current (DC)-DC converters. In some examples, the power supply devices **102** include point-of-load (POL) voltage regulators.
(19) As shown in FIG. **1**B, the power supply system **100** may include a power supply device **102-1**, a power supply device **102-2**, and a power supply device **102-3** through power supply device **102-**N. It is noted that the power supply system **100** may include any number of power supply devices **102** such as two, three, four, or any number greater than four. A power supply device **102** may refer to any of the power supply devices **102-1** through **102-**N. Each power supply device **102** has an output (e.g., a terminal, pin, etc.) configured to provide a respective rail voltage **106** to a connected device (e.g., the integrated circuit **104**). As shown in FIG. **1**B, the power supply device **102-1** generates a rail voltage **106-1**, the power supply device **102-2** generates a rail voltage **106-2**, the power supply device **102-3** generates a rail voltage **106-3**, and the power supply device **102-**N generates a rail voltage **106-**N.
(20) Each power supply device **102** may receive an enable signal **112** as an input. The enable signal **112** may be a common enable signal between the power supply devices **102**. For example, each power supply device **102** may receive the enable signal **112** (e.g., the same enable signal). Activation of the enable signal **112** causes the power supply devices **102** to be activated according to an activation power sequence. Activation of the enable signal **112** may include a transition from a first voltage level to a second voltage level (e.g., a rising edge). The activation power sequence may indicate that power supply device **102-1** is activated first, and, after the rail voltage **106-1** has reached its target value, the power supply device **102-2** is activated, which is followed by the

activation of power supply device **102-3**, and so forth. Deactivation of the enable signal **112** causes the power supply devices **102** to be deactivated according to a deactivation power sequence. Deactivation of the enable signal **112** may include a transition from the second voltage level to the first voltage level (e.g., a falling edge). The deactivation power sequence may indicate that power supply device **102-1** is deactivated first, and, after the rail voltage **106-1** has reached its minimum value, the power supply device **102-2** is deactivated, which is followed by the deactivation of power supply device **102-3**, and so forth. In some examples, the deactivation power sequence has a timing order that is the same as the activation power sequence. In some examples, the deactivation power sequence has a timing order that is different from the activation power sequence.

(21) When the enable signal **112** is activated, the power supply devices **102** detect the transition (e.g., the rising edge) of the enable signal **112**, and a sequence delay counter **120**, included within each power supply device **102**, controls when a respective rail voltage **106** is activated (e.g., starts to increase to its target voltage level). When the enable signal **112** is deactivated, the power supply devices **102** detect the transition (e.g., the falling edge) of the enable signal **112**, and the sequence delay counter **120**, included within each power supply device **102**, controls when a respective rail voltage **106** is deactivated (e.g., starts to decrease to a minimum voltage).

(22) Each power supply device **102** is configured to receive a clock signal **114** as an input. The clock signal **114** may be a common clock signal for the power supply devices **102**. Each power supply device **102** may include an interrupt output (e.g., a terminal, pin, etc.) that provides the interrupt signal **110**. The integrated circuit **104** may receive the interrupt signal **110**. In some examples, the interrupt signal **110** is a signal common to the power supply device **102**. For example, activation of the interrupt signal **110** by any one of the power supply devices **102** may cause the integrated circuit **104** to operate in a safe state. Activation of the interrupt signal **110** may include transitioning from a first voltage level to a second voltage level (e.g., a rising edge). A safe state may be an operating state of the integrated circuit **104**. The integrated circuit **104** may operate in a normal operating state when the integrated circuit **104** is functioning normally. Activation of the interrupt signal **110** may cause the integrated circuit **104** to change operating states from a first operating state (e.g., a normal operating state) to a second operating state (e.g., a safe operating state). The integrated circuit **104** in the second operating state may function differently than the integrated circuit **104** in the first operating state.

(23) Each power supply device **102** may activate an interrupt signal **110** in response to the detection of a rail violation **134**. The power supply system **100** may include a communication bus **108** connected to the integrated circuit **104** and the power supply devices **102**. In some examples, the communication bus **108** includes an inter-integrated circuit (I2C) bus.

(24) FIG. **1**C illustrates an example of a power supply device **102**, which may be any of the power supply devices **102** in the power supply system **100**. As shown in FIG. **1**C, the power supply device **102** includes a sequence delay counter **120** and a voltage regulator **122**. The voltage regulator **122** is configured to generate the rail voltage **106** from the supply voltage **116**. The sequence delay counter **120** may receive the enable signal **112** and the clock signal **114** and control the timing of when the voltage regulator **122** activates according to the activation power sequence and deactivates according to the deactivation power sequence.

(25) The sequence delay counter **120** may trigger activation (or deactivation) of the voltage regulator **122** based on counter values **124**. The counter values may include a power-up delay value **126** and a power-down delay value **128**. The power-up delay value **126** may be used during the activation power sequence. The power-down delay value **128** may be used during the deactivation power sequence. The power-up delay value **126** may represent a length of time that the voltage regulator **122** is delayed until the voltage regulator **122** is activated. When the voltage regulator **122** is activated, the rail voltage **106** starts to increase. The power-up delay value **126** may indicate a period of time from detection of the activation of the enable signal **112**. In other words, instead of activating the voltage regulator **122** when the activation of the enable signal **112** is detected, the

activation of the voltage regulator **122** is delayed by the power-up delay value **126**. In response to detecting that the enable signal **112** is activated, the sequence delay counter **120** may delay activation of the voltage regulator **122** according to the power-up delay value **126**. When the sequence delay counter **120** triggers activation of the voltage regulator **122**, the rail voltage **106** increases to its target value (e.g., to be powered up).

(26) The power-down delay value **128** may represent a length of time that the voltage regulator **122** is delayed until the voltage regulator **122** is deactivated. When the voltage regulator **122** is deactivated, the rail voltage **106** starts to decrease. The power-down delay value **128** may indicate a period of time from detection of the deactivation of the enable signal **112**. In other words, instead of deactivating the voltage regulator **122** when the deactivation of the enable signal **112** is detected, the deactivation of the voltage regulator **122** is delayed by the power-down delay value **128**. In response to detecting that the enable signal **112** is deactivated, the sequence delay counter **120** may delay deactivation of the voltage regulator **122** according to the power-down delay value **128**. For example, the sequence delay counter **120** may trigger deactivation of the voltage regulator **122** in response to the period of time (starting from when the deactivation of the enable signal **112** is detected) being equal to (or greater than) the power-down delay value **128**, which causes the rail voltage **106** to decrease from its target value (e.g., to be powered down).

(27) In some examples, the power supply device **102** includes a memory device **160** configured to store the counter values **124**. In some examples, the memory device **160** includes a field programmable memory device. In some examples, the memory device **160** includes a one-time programmable (OTP) memory device. In some examples, the power supply device **102** does not include a separate memory device, and the counter values **124** are stored at the sequence delay counter **120**.

(28) Referring to FIGS. **1**C and **1**D, the power supply device **102** includes an internal diagnostic circuit **130** configured to detect a rail violation **134** with respect to its rail voltage **106** during the power sequence (e.g., the activation power sequence, the deactivation power sequence), and, when detected, may activate an interrupt signal **110**, which causes the integrated circuit **104** to operate in a safe state. For example, when the enable signal **112** is activated (or deactivated), the internal diagnostic circuit **130** may detect whether its respective rail voltage **106** has started-up too early or too late in the power sequence by monitoring the rail voltage **106** at one or more times during a monitoring window **142**.

(29) In response to activation of the enable signal **112**, the internal diagnostic circuit **130** may detect whether a value of the rail voltage **106** satisfies a first condition defined by a voltage threshold **150-1** at a first time **141-1** (e.g., the start of the monitoring window **142**) to ensure that the power supply device **102** has not powered up too early. The voltage threshold **150-1** may be an undervoltage threshold, and the first condition defined by the voltage threshold **150-1** may be that the value of the rail voltage **106** is less than the voltage threshold **150-1** at the first time **141-1**. The value of the rail voltage **106** being equal to or greater than the voltage threshold **150-1** at the first time **141-1** may indicate a rail violation **134** (e.g., powered up too early).

(30) During activation, the internal diagnostic circuit **130** may detect whether the value of the rail voltage **106** satisfies a second condition defined by the voltage threshold **150-1** at a second time **141-2** (e.g., the end of the monitoring window **142**) to ensure that the power supply device **102** has not powered up too late. The second condition of the voltage threshold **150-1** may be that the value of the rail voltage **106** is greater than the voltage threshold **150-1** at the second time **141-2**. The value of the rail voltage **106** being equal to or less than the voltage threshold **150-1** at the second time **141-2** may indicate a rail violation **134** (e.g., powered up too late). In some examples, after a power supply device **102** has been activated (e.g., powered-up), the internal diagnostic circuit **130** may monitor the rail voltage **106** to ensure that the rail voltage **106** is greater than the voltage threshold **150-1**. For example, after the power supply device **102** has been activated, the internal diagnostic circuit **130** may detect whether the rail voltage **106** is greater than the voltage threshold

150-1, and, if not, may activate the interrupt signal **110**.
(31) In response to deactivation of the enable signal **112**, the internal diagnostic circuit **130** may detect whether the value of the rail voltage **106** satisfies a first condition defined by a voltage threshold **150-2** at a first time **141-1** (e.g., the start of the monitoring window **142**) to ensure that the power supply device **102** has not powered down too early. In some examples, the voltage threshold **150-2** is a lower threshold (e.g., a disabled threshold). The first condition defined by the voltage threshold **150-2** may be that the value of the rail voltage **106** is greater than the voltage threshold **150-2**. The value of the rail voltage **106** being equal to or less than the voltage threshold **150-2** at the first time **141-1** may indicate a rail violation **134** (e.g., powered down too early).
(32) During deactivation, the internal diagnostic circuit **130** may detect whether the value of the rail voltage **106** satisfies a second condition defined by the voltage threshold **150-2** at a second time **141-2** (e.g., the end of the monitoring window **142**) to ensure that the power supply device **102** has not powered down too late. The second condition defined by the voltage threshold **150-2** may be that the value of the rail voltage **106** is less than or equal to the voltage threshold **150-2** at the second time **141-2**. The value of the rail voltage **106** being equal to or greater than the voltage threshold **150-2** at the second time **141-2** may indicate a rail violation **134** (e.g., powered down too late).
(33) The internal diagnostic circuit **130** includes a sequence delay check counter **132** and a voltage threshold detector **146**. The sequence delay check counter **132** may determine when to check for a rail violation **134** (e.g., at the first time **141-1**, the second time **141-2**) based on counter values **135** associated with the sequence delay check counter **132**. The voltage threshold detector **146** may detect whether the value of the rail voltage **106** satisfies a condition (e.g., the first condition, second condition) (e.g., greater, less than, and/or equal to) defined by one or more voltage thresholds **150** (e.g., the voltage threshold **150-1** or the voltage threshold **150-1**). In some examples, the voltage threshold detector **146** may activate or deactivate a voltage threshold signal **170** and/or a voltage threshold signal **172** based on a comparison of the value of the rail voltage **106** with the voltage threshold **150-1** and the voltage threshold **150-2**.
(34) In further detail, the voltage threshold detector **146** may detect whether the value of the rail voltage **106** is less than a voltage threshold **150-1**, and, if so, may activate a voltage threshold signal **170**. In some examples, the voltage threshold **150-1** is an undervoltage threshold. In some examples, the voltage threshold **150-1** is used during activation of the power supply device **102**. If the voltage threshold detector **146** detects that the value of the rail voltage **106** is equal to or greater than the voltage threshold **150-1**, the voltage threshold detector **146** may deactivate the voltage threshold signal **170**. The voltage threshold detector **146** may detect whether the value of the rail voltage **106** is less than the voltage threshold **150-2**, and, if so, may activate a voltage threshold signal **172**. In some examples, the voltage threshold **150-2** is used during deactivation of the power supply device **102**. In some examples, the voltage threshold **150-2** is less than the voltage threshold **150-1**. If the voltage threshold detector **146** detects that the rail voltage **106** is equal to or greater than the voltage threshold **150-2**, the voltage threshold detector **146** may deactivate the voltage threshold signal **172**.
(35) The sequence delay check counter **132** may receive the enable signal **112**, the clock signal **114**, and, in some examples, the voltage threshold signals **170**, **172** from the voltage threshold detector **146**. The sequence delay check counter **132** may detect a rail violation **134** at the first time **141-1** and the second time **141-2** defined by the counter values **135** using the voltage threshold signal **170** and the voltage threshold signal **172**. The counter values **135** may include a power-up check value **136**, a power-down check value **138**, and an error window value **140**. The power-up check value **136** may indicate a first time (e.g., the first time **141-1**) from activation of the enable signal **112** to check for a rail violation **134**. During activation, the error window value **140** may represent a second time (e.g., the second time **141-2**) from the power-up check value **136** to check for a rail violation **134**. The power-down check value **138** may indicate a first time (e.g., the first time **141-1**)

from deactivation of the enable signal **112** to check for a rail violation **134**. During deactivation, the error window value **140** may also represent a second time (e.g., the second time **141-2**) from the power-down check value **138** to check for a rail violation **134**. In some examples, the error window value **140** includes a first error window value indicating a length of time during activation, and a second error window value indicating a length of time during deactivation. In some examples, the second error window value is different from the first error window value. In some examples, the second error window value is the same as the first error window value.

(36) Referring to FIG. **1**E, in response to activation of the enable signal **112**, the sequence delay counter **120** causes the activation of the voltage regulator **122** to be delayed according to the power-up delay value **126**. When the voltage regulator **122** is activated, the rail voltage **106** starts to increase.

(37) In response to activation of the enable signal **112**, the sequence delay check counter **132** detects, at a time (e.g., the first time **141-1**) indicated by the power-up check value **136**, whether or not the voltage threshold signal **170** is activated. In other words, when a period of time starting from the receipt of the enable signal **112** has reached the power-up check value **136**, the sequence delay check counter **132** may detect whether or not the voltage threshold signal **170** is activated. If the voltage threshold signal **170** is activated (e.g., which indicates that the value of the rail voltage **106** is less than the voltage threshold **150-1**), the sequence delay check counter **132** may determine that the activation power sequence is operating normally (e.g., a rail violation **134** is not detected). If the voltage threshold signal **170** is deactivated (e.g., which indicates that the value of the rail voltage **106** is equal to or greater than the voltage threshold **150-1**), the sequence delay check counter **132** may detect a rail violation **134** and activate the interrupt signal **110**.

(38) The sequence delay check counter **132** may detect, at a time (e.g., the second time **141-2**) defined by the error window value **140**, whether or not the voltage threshold signal **170** is activated. In other words, when a period of time starting from the time represented by the power-up check value **136** has reached the error window value **140**, the sequence delay check counter **132** may detect whether or not the voltage threshold signal **170** is activated. If the voltage threshold signal **170** is deactivated (e.g., which indicates that the value of the rail voltage **106** is equal to or greater than the voltage threshold **150-1**), the sequence delay check counter **132** may determine that the activation power sequence is operating normally (e.g., a rail violation **134** is not detected). If the voltage threshold signal **170** is activated (e.g., which indicates that the value of the rail voltage **106** is less than the voltage threshold **150-1**), the sequence delay check counter **132** may detect a rail violation **134** and activate the interrupt signal **110**.

(39) In response to deactivation of the enable signal **112**, the sequence delay counter **120** causes the deactivation of the voltage regulator **122** to be delayed according to the power-down delay value **128**. When the voltage regulator **122** is deactivated, the rail voltage **106** starts to decrease.

(40) In response to deactivation of the enable signal **112**, the sequence delay check counter **132** detects, at a time (e.g., the first time **141-1**) defined by the power-down check value **138**, whether or not the voltage threshold signal **172** is activated. In other words, when a period of time starting from the receipt of deactivation of the enable signal **112** has reached the power-down check value **138**, the sequence delay check counter **132** may detect whether or not the voltage threshold signal **172** is activated. If the voltage threshold signal **172** is deactivated (e.g., which indicates that the value of the rail voltage **106** is equal to or greater than the voltage threshold **150-2**), the sequence delay check counter **132** may determine that the deactivation power sequence is operating normally (e.g., a rail violation **134** is not detected). If the voltage threshold signal **172** is activated (e.g., which indicates that the value of the rail voltage **106** is less than the voltage threshold **150-2**), the sequence delay check counter **132** may detect a rail violation **134** and activate the interrupt signal **110**.

(41) During deactivation, the sequence delay check counter **132** may detect, at a time (e.g., the second time **141-2**) defined by the error window value **140**, whether or not the voltage threshold

signal **172** is activated. In other words, when a period of time starting from the time represented by the power-down check value **138** has reached the error window value **140**, the sequence delay check counter **132** may detect whether or not the voltage threshold signal **172** is activated. If the voltage threshold signal **172** is activated (e.g., which indicates that the value of the rail voltage **106** is less than the voltage threshold **150-2**), the sequence delay check counter **132** may determine that the deactivation power sequence is operating normally (e.g., a rail violation **134** is not detected). If the voltage threshold signal **172** is deactivated (e.g., which indicates that the value of the rail voltage **106** is greater than or equal to the voltage threshold **150-2**), the sequence delay check counter **132** may detect a rail violation **134** and activate the interrupt signal **110**.
(42) In some examples, the counter values **124** used by the sequence delay counter **120** and/or the counter values **135** used by the sequence delay check counter **132** are field programmable. For example, the sequence delay counter **120** and the sequence delay check counter **132** may be programmable after the power supply device **102** has been manufactured. The memory device **160** may receive programming data **182**, where the programming data **182** includes the counter values **124** and/or the counter values **135**. In some examples, the programming data **182** may include other configuration data such as the target voltage levels, the power-down voltage levels, the voltage threshold **150-1**, and/or the voltage threshold **150-2**. The counter values **124** and/or the counter values **135** from the programming data **182** may be stored in the memory device **160** and/or in the sequence delay counter **120**, the sequence delay check counter **132**, and/or the voltage regulator **122**. In some examples, the integrated circuit **104** may transmit the programming data **182** to the power supply devices **102**. In some examples, the integrated circuit **104** transmits the programming data **182** via the communication bus **108**. In some examples, a controller **180**, separate from the integrated circuit **104**, may provide the programming data **182** to the power supply devices **102**.
(43) In some examples, each power supply device **102** is associated with an identifier that can uniquely identify a respective power supply device **102**, and the programming data **182** may include the identifier for a power supply device **102** to be programmed with the counter values **124**. In some examples, the identifier is an address that identifies the power supply device **102**. In some examples, the programming data **182** includes digital data. In some examples, the programming data **182** includes analog data. In some examples, the addressing via a resistor and an analog-to-digital converter.
(44) Referring to FIG. **1**F, in response to activation of the enable signal **112**, the rail voltage **106-1** is activated at a time represented by a power-up delay value **126-1**. The power supply device **102-1** checks for a rail violation **134** on the rail voltage **106-1** during a monitoring window **142*a*-1** (e.g., at a first time **141-1** and a second time **141-2**). Then, the rail voltage **106-2** is activated at a time represented by a power-up delay value **126-2**. The power supply device **102-2** checks for a rail violation **134** on the rail voltage **106-2** during a monitoring window **142*a*-2** (e.g., at a first time **141-1** and a second time **141-2**). Then, the rail voltage **106-3** is activated at a time represented by a power-up delay value **126-3**. The power supply device **102-3** detects a rail violation **134** on the rail voltage **106-3** during a monitoring window **142*a*-3** (e.g., at a first time **141-1** and a second time **141-2**) and activates the interrupt signal **110**. In some examples, after a power supply device **102** has been activated (e.g., powered-up), the internal diagnostic circuit **130** may monitor its respective rail voltage **106** (e.g., **106-1**, **106-2**, or **106-3**) to ensure that the rail voltage **106** is greater than the voltage threshold **150-1**. For example, after the power supply device **102-1** has been activated, the internal diagnostic circuit **130** may detect whether the rail voltage **106-1** is greater than the voltage threshold **150-1** during a period **139** (e.g., at one or more times (or continuous monitoring) during the period **139** between the monitoring window **142*a*-1** and the monitoring window **142*b*-1**), and, if not, may activate the interrupt signal **110**. In some examples, the period **139** may be between the second time **141-2** associated with the monitoring window **142*a*-1** and the first time **141-1** associated with the monitoring window **142*b*-1**.
(45) In response to deactivation of the enable signal **112**, the rail voltage **106-1** is deactivated at a

time represented by a power-down delay value **128-1**. The power supply device **102-1** checks for a rail violation **134** on the rail voltage **106-1** during a monitoring window **142**_b_**-1** (e.g., at a first time **141-1** and a second time **141-2**). Then, the rail voltage **106-3** is deactivated at a time represented by a power-down delay value **128-3**. The power supply device **102-3** checks for a rail violation **134** on the rail voltage **106-3** during a monitoring window **142**_b_**-3** (e.g., at a first time **141-1** and a second time **141-2**). Then, the rail voltage **106-2** is deactivated at a time represented by a power-up delay value **128-2**. The power supply device **102-2** checks for a rail violation **134** on the rail voltage **106-2** during a monitoring window **142**_b_**-2** (e.g., at a first time **141-1** and a second time **141-2**). In some examples, the monitoring window **142**_b_**-1**, **142**_b_**-2**, or **142**_b_**-3** has a length that is different from a corresponding monitoring window **142**_a_**-1**, **142**_a_**-2**, or **142**_a_**-3**. In some examples, the monitoring window **142**_b_**-1**, **142**_b_**-2**, or **142**_b_**-3** has a length that is the same as a corresponding monitoring window **142**_a_**-1**, **142**_a_**-2**, or **142**_a_**-3**.

(46) FIG. **2** illustrates an example of a power supply device **202** according to an aspect. The power supply device **202** may be an example of the power supply device **102** of FIGS. **1**A to **1**F and may include any of the details discussed with reference to those figures. The power supply device **202** includes a sequence delay counter **220**, a voltage regulator **222**, an internal diagnostic circuit **230**, and a memory device **260**. The sequence delay counter **220** may receive a clock signal **214** and an enable signal **212**, and delay activation (or deactivation) of the voltage regulator **222** according to its counter values. In some examples, the power supply device **202** includes a communication bus **208**. The voltage regulator **222** is configured to generate a rail voltage **206**. The internal diagnostic circuit **230** may receive the enable signal **212** (and the clock signal **214**), and the rail voltage **206**, and detect a rail violation. If a rail violation is detected, the internal diagnostic circuit **230** may activate an interrupt signal **210**. The internal diagnostic circuit **230** includes a sequence delay check counter **232** configured to receive the enable signal **212**, a voltage threshold signal **270**, and a voltage threshold signal **272**.

(47) The memory device **260** may receive programming data **282**, where the programming data **282** includes counter values **224** used by the sequence delay counter **220** and/or counter values **235** used by the sequence delay check counter **232**. In some examples, the programming data **282** may include other configuration data such as the target voltage levels, the power-down voltage levels, voltage thresholds (e.g., the voltage threshold **150-1**, the voltage threshold **150-2**). In some examples, an integrated circuit may transmit the programming data **282** to the power supply devices **202**. In some examples, the integrated circuit **104** transmits the programming data **282** via the communication bus **208**. In some examples, the communication bus **208** is an I2C bus. In some examples, the integrated circuit **104** transmits the programming data **282** via the communication bus **208**. In some examples, the power supply device **202** is associated with an address **292** that can uniquely identify the power supply device **202**, and the programming data **282** may include the address **292** with the counter values **224** and/or the counter values **235**.

(48) FIG. **3** illustrates an example of a power supply device **302** according to an aspect. The power supply device **302** may be an example of the power supply device **102** of FIGS. **1**A to **1**F and may include any of the details discussed with reference to those figures. The power supply device **302** includes a sequence delay counter **320**, a voltage regulator **322**, an internal diagnostic circuit **330**, and a memory device **360**. The sequence delay counter **320** may receive a clock signal **314** and an enable signal **312**, and delay activation (or deactivation) of the voltage regulator **322** according to its counter values. In some examples, the power supply device **302** includes a communication bus **308**. The voltage regulator **322** is configured to generate a rail voltage **306**. The internal diagnostic circuit **330** may receive the enable signal **312** (and the clock signal **314**), and the rail voltage **306**, and detect a rail violation. If a rail violation is detected, the internal diagnostic circuit **330** may activate an interrupt signal **310**. The internal diagnostic circuit **330** includes a sequence delay check counter **332** configured to receive the enable signal **312**, a voltage threshold signal **370**, and a voltage threshold signal **372**.

(49) The memory device **360** may receive programming data **382**, where the programming data **382** includes counter values **324** used by the sequence delay counter **320** and/or counter values **335** used by the sequence delay check counter **332**. In some examples, the programming data **382** may include other configuration data such as the target voltage levels, the power-down voltage levels, voltage thresholds (e.g., the voltage threshold **150-1**, the voltage threshold **150-2**). In some examples, the power supply device **302** is associated with an address **392** that can uniquely identify the power supply device **302**. In some examples, the power supply device **302** includes an analog-to-digital converter **362** and a resistor **364**. In some examples, the address **392** is defined by the analog-to-digital converter **362** and the resistor **364**. The power supply device **302** may drive a current into the resistor **364**, which creates a voltage that is monitored by the analog-to-digital converter **362**. The analog-to-digital converter **362** may obtain a digital code and compare the digital code with hardcoded codes to define the address **392** of the power supply device **302**. In some examples, changing the resistor value of the resistor **364** may change the voltage, the digital code, and therefore the address **392** of the power supply device **302**.

(50) FIG. **4** illustrates a flowchart **400** depicting example operations of a power supply system according to an aspect. Although the flowchart **400** is explained with reference to the power supply system **100** of FIGS. **1**A through **1**F, the operations may be implemented by any of the examples discussed herein. Although the flowchart **400** of FIG. **4** illustrates the operations in sequential order, it will be appreciated that this is merely an example, and that additional or alternative operations may be included. Further, operations of FIG. **4** and related operations may be executed in a different order than that shown, or in a parallel or overlapping fashion.

(51) Operation **402** includes, in response to activation of an enable signal to activate the power system according to an activation power sequence, activating a power supply device to generate a rail voltage. Operation **404** includes detecting a rail violation of the activation power sequence in response to a value of the rail voltage, at a first time, not satisfying a first condition defined by a voltage threshold, or the value of the rail voltage, at a second time, not satisfying a second condition defined by the voltage threshold. Operation **406** includes, in response to the rail violation being detected, activating an interrupt signal, the interrupt signal configured to cause an integrated circuit to operate in a safe state.

(52) In some aspects, the techniques described herein relate to a power supply system, including: a plurality of power supply devices configured to be connected to an integrated circuit, a power supply device of the plurality of power supply devices including: a voltage regulator configured to generate a rail voltage; and an internal diagnostic circuit configured to: detect activation of an enable signal to activate the plurality of power supply devices according to an activation power sequence; detect a rail violation of the activation power sequence in response to a value of the rail voltage: at a first time, not satisfying a first condition defined by a voltage threshold, or at a second time, not satisfying a second condition defined by the voltage threshold; and in response to the rail violation being detected, activate an interrupt signal.

(53) In some aspects, the techniques described herein relate to a power supply system, wherein the first condition defined by the voltage threshold includes the value of the rail voltage being less than the voltage threshold. In some aspects, the techniques described herein relate to a power supply system, wherein the second condition defined by the voltage threshold includes the value of the rail voltage being equal to or greater than the voltage threshold.

(54) In some aspects, the techniques described herein relate to a power supply system, wherein the voltage threshold is a first voltage threshold, the internal diagnostic circuit is configured to: detect deactivation of the enable signal to deactivate the plurality of power supply devices according to a deactivation power sequence; and detect a rail violation of the deactivation power sequence in response to the value of the rail voltage: at a third time, not satisfying a first condition defined by a second voltage threshold, or at a fourth time, not satisfying a second condition defined by the second voltage threshold.

(55) In some aspects, the techniques described herein relate to a power supply system, wherein the first condition defined by the second voltage threshold includes the value of the rail voltage being equal to or greater than the second voltage threshold. In some aspects, the techniques described herein relate to a power supply system, wherein the second condition defined by the second voltage threshold includes the rail voltage being less than the second voltage threshold.
(56) In some aspects, the techniques described herein relate to a power supply system, wherein the internal diagnostic circuit includes a sequence delay check counter associated with a plurality of counter values, the plurality of counter values including a first counter value corresponding to the first time and a second counter value corresponding to the second time.
(57) In some aspects, the techniques described herein relate to a power supply system, wherein the internal diagnostic circuit includes a voltage threshold detector configured to activate or deactivate a voltage threshold signal based on a comparison of the value of the rail voltage with the voltage threshold, wherein the sequence delay check counter is configured to detect the rail violation based on whether the voltage threshold signal is activated or deactivated at the first time or the second time.
(58) In some aspects, the techniques described herein relate to a power supply system, wherein the internal diagnostic circuit is configured to receive programming data to configure the sequence delay check counter, the programming data including the plurality of counter values. In some aspects, the techniques described herein relate to a power supply system, wherein the internal diagnostic circuit includes a memory device configured to store the plurality of counter values. In some aspects, the techniques described herein relate to a power supply system, wherein the power supply device is associated with an address, the internal diagnostic circuit including an analog-to-digital converter and a resistor, the analog-to-digital converter and the resistor configured to define the address.
(59) In some aspects, the techniques described herein relate to a power supply device including: a voltage regulator configured to generate a rail voltage; and an internal diagnostic circuit configured to: detect activation of an enable signal to activate a plurality of power supply devices according to an activation power sequence; detect a rail violation of the activation power sequence in response to a value of the rail voltage: at a first time, not satisfying a first condition defined by a voltage threshold, or at a second time, not satisfying a second condition defined by the voltage threshold; and in response to the rail violation, activate an interrupt signal. In some aspects, the techniques described herein relate to a power supply device, wherein the first condition defined by the voltage threshold includes the value of the rail voltage being less than the voltage threshold, wherein the second condition defined by the voltage threshold includes the value of the rail voltage being equal to or greater than the voltage threshold.
(60) In some aspects, the techniques described herein relate to a power supply device, wherein the voltage threshold is a first voltage threshold, the internal diagnostic circuit is configured to: detect deactivation of the enable signal to deactivate the plurality of power supply devices according to a deactivation power sequence; and detect a rail violation of the deactivation power sequence in response to the value of the rail voltage: at a third time, not satisfying a first condition defined by a second voltage threshold, or at a fourth time, not satisfying a second condition defined by the second voltage threshold.
(61) In some aspects, the techniques described herein relate to a power supply device, wherein the internal diagnostic circuit includes: a sequence delay check counter associated with a plurality of counter values that correspond to the first through fourth times. In some aspects, the techniques described herein relate to a power supply device, wherein the first condition defined by the second voltage threshold includes the value of the rail voltage being equal to or greater than the second voltage threshold, wherein the second condition defined by the second voltage threshold includes the rail voltage being less than the second voltage threshold.
(62) In some aspects, the techniques described herein relate to a method of monitoring a power

sequence of a power supply system, the method including: in response to activation of an enable signal to activate a power supply system according to an activation power sequence, activating a power supply device to generate a rail voltage; detecting a rail violation of the activation power sequence in response to a value of the rail voltage: at a first time, not satisfying a first condition defined by a voltage threshold, or at a second time, not satisfying a second condition defined by the voltage threshold; and in response to the rail violation being detected, activating an interrupt signal.
(63) In some aspects, the techniques described herein relate to a method, wherein the first condition defined by the voltage threshold includes the value of the rail voltage being less than the voltage threshold, wherein the second condition defined by the voltage threshold includes the rail voltage being greater than or equal to the voltage threshold.
(64) In some aspects, the techniques described herein relate to a method, wherein the voltage threshold is a first voltage threshold, the method further including: detecting deactivation of the enable signal to deactivate the power supply system according to a deactivation power sequence; and detecting a rail violation of the deactivation power sequence in response to the value of the rail voltage: at a third time, not satisfying a first condition defined by a second voltage threshold, or at a fourth time, not satisfying a second condition defined by the second voltage threshold.
(65) In some aspects, the techniques described herein relate to a method, further including: detecting, by a sequence delay check counter, activation of the enable signal; triggering, by the sequence delay check counter, a rail check in response to a period of time satisfying a counter value that represents the first time; and in response to the rail check, detecting whether the value of the rail voltage satisfies the first condition defined by the voltage threshold.
(66) Various implementations of the systems and techniques described here can be realized in digital electronic circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device. Various implementations of the systems and techniques described here can be realized as and/or generally be referred to herein as a circuit, a module, a block, or a system that can combine software and hardware aspects. For example, a module may include the functions/acts/computer program instructions executing on a processor (e.g., a processor formed on a silicon substrate, a GaAs substrate, and the like) or some other programmable data processing apparatus.
(67) Some of the above examples are described as processes or methods depicted as flowcharts. Although the flowcharts describe the operations as sequential processes, many of the operations may be performed in parallel, concurrently or simultaneously. In addition, the order of operations may be re-arranged. The processes may be terminated when their operations are completed but may also have additional steps not included in the figure. The processes may correspond to methods, functions, procedures, subroutines, subprograms, etc.
(68) Methods discussed above, some of which are illustrated by the flow charts, may be implemented by hardware, software, firmware, middleware, microcode, hardware description languages, or any combination thereof. When implemented in software, firmware, middleware or microcode, the program code or code segments to perform the necessary tasks may be stored in a machine or computer readable medium such as a storage medium. A processor(s) may perform the necessary tasks.
(69) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from

the scope of examples discussed herein. As used herein, the term and/or includes any and all combinations of one or more of the associated listed items.

(70) It will be understood that, in the foregoing description, when an element is referred to as being on, connected to, electrically connected to, coupled to, or electrically coupled to another element, it may be directly on, connected or coupled to the other element, or one or more intervening elements may be present. In contrast, when an element is referred to as being directly on, directly connected to or directly coupled to another element, there are no intervening elements present. Although the terms directly on, directly connected to, or directly coupled to may not be used throughout the detailed description, elements that are shown as being directly on, directly connected or directly coupled can be referred to as such. The claims of the application, if any, may be amended to recite exemplary relationships described in the specification or shown in the figures.

(71) As used in this specification, a singular form may, unless definitely indicating a particular case in terms of the context, include a plural form. Spatially relative terms (e.g., over, above, upper, under, beneath, below, lower, and so forth) are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. In some implementations, the relative terms above and below can, respectively, include vertically above and vertically below. In some implementations, the term adjacent can include laterally adjacent to or horizontally adjacent to.

(72) While certain features of the described implementations have been illustrated as described herein, many modifications, substitutions, changes and equivalents will now occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the scope of the implementations. It should be understood that they have been presented by way of example only, not limitation, and various changes in form and details may be made. Any portion of the apparatus and/or methods described herein may be combined in any combination, except mutually exclusive combinations. The implementations described herein can include various combinations and/or sub-combinations of the functions, components and/or features of the different implementations described.

(73) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, e.g., those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(74) Portions of the above example embodiments and corresponding detailed description are presented in terms of software, or algorithms and symbolic representations of operation on data bits within a computer memory. These descriptions and representations are the ones by which those of ordinary skill in the art effectively convey the substance of their work to others of ordinary skill in the art. An algorithm, as the term is used here, and as it is used generally, is conceived to be a self-consistent sequence of steps leading to a desired result. The steps are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of optical, electrical, or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

(75) In the above illustrative embodiments, reference to acts and symbolic representations of operations (e.g., in the form of flowcharts) that may be implemented as program modules or functional processes include routines, programs, objects, components, data structures, etc., that perform particular tasks or implement particular abstract data types and may be described and/or implemented using existing hardware at existing structural elements. Such existing hardware may include one or more Central Processing Units (CPUs), digital signal processors (DSPs),

application-specific-integrated-circuits, field programmable gate arrays (FPGAs) computers or the like.
(76) It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise, or as is apparent from the discussion, terms such as processing or computing or calculating or determining of displaying or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical, electronic quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.
(77) Lastly, it should also be noted that whilst the accompanying claims set out particular combinations of features described herein, the scope of the present disclosure is not limited to the particular combinations hereafter claimed, but instead extends to encompass any combination of features or embodiments herein disclosed irrespective of whether or not that particular combination has been specifically enumerated in the accompanying claims at this time.

Claims

1. A power supply system, comprising: a plurality of power supply devices configured to be connected to an integrated circuit, a power supply device of the plurality of power supply devices including: a voltage regulator configured to generate a rail voltage; and an internal diagnostic circuit configured to: detect activation of an enable signal to activate the plurality of power supply devices according to an activation power sequence; detect a rail violation of the activation power sequence in response to a value of the rail voltage: at a first time represented by a first counter value included in programming data, not satisfying a first condition defined by a voltage threshold, or at a second time represented by a second counter value, not satisfying a second condition defined by the voltage threshold; and in response to the rail violation being detected, activate an interrupt signal.
2. The power supply system of claim 1, wherein the first condition defined by the voltage threshold includes the value of the rail voltage being less than the voltage threshold.
3. The power supply system of claim 1, wherein the second condition defined by the voltage threshold includes the value of the rail voltage being equal to or greater than the voltage threshold.
4. The power supply system of claim 1, wherein the voltage threshold is a first voltage threshold, and wherein the internal diagnostic circuit is configured to: detect deactivation of the enable signal to deactivate the plurality of power supply devices according to a deactivation power sequence; and detect a rail violation of the deactivation power sequence in response to the value of the rail voltage: at a third time, not satisfying a first condition defined by a second voltage threshold, or at a fourth time, not satisfying a second condition defined by the second voltage threshold.
5. The power supply system of claim 4, wherein the first condition defined by the second voltage threshold includes the value of the rail voltage being equal to or greater than the second voltage threshold.
6. The power supply system of claim 4, wherein the second condition defined by the second voltage threshold includes the rail voltage being less than the second voltage threshold.
7. The power supply system of claim 1, wherein the internal diagnostic circuit includes a sequence delay check counter being configured with a plurality of counter values, the plurality of counter values including the first counter value and the second counter value.
8. The power supply system of claim 7, wherein the internal diagnostic circuit includes a voltage threshold detector configured to activate or deactivate a voltage threshold signal based on a comparison of the value of the rail voltage with the voltage threshold, and wherein the sequence delay check counter is configured to detect the rail violation based on whether the voltage threshold

signal is activated or deactivated at the first time or the second time.
9. The power supply system of claim 7, wherein the internal diagnostic circuit is configured to receive the programming data to configure the sequence delay check counter, the programming data including the plurality of counter values.
10. The power supply system of claim 9, wherein the internal diagnostic circuit includes a memory device configured to store the plurality of counter values.
11. The power supply system of claim 9, wherein the power supply device is associated with an address, the internal diagnostic circuit including an analog-to-digital converter and a resistor, the analog-to-digital converter and the resistor configured to define the address.
12. A power supply device comprising: a voltage regulator configured to generate a rail voltage; and an internal diagnostic circuit configured to: detect activation of an enable signal to activate a plurality of power supply devices according to an activation power sequence, the power supply device not using a power management integrated circuit for controlling a power sequence of the plurality of power supply devices; detect a rail violation of the activation power sequence in response to a value of the rail voltage: at a first time, not satisfying a first condition defined by a voltage threshold, or at a second time, not satisfying a second condition defined by the voltage threshold; and in response to the rail violation, activate an interrupt signal.
13. The power supply device of claim 12, wherein the first condition defined by the voltage threshold includes the value of the rail voltage being less than the voltage threshold, and wherein the second condition defined by the voltage threshold includes the value of the rail voltage being equal to or greater than the voltage threshold.
14. The power supply device of claim 12, wherein the voltage threshold is a first voltage threshold, the internal diagnostic circuit is configured to: detect deactivation of the enable signal to deactivate the plurality of power supply devices according to a deactivation power sequence; and detect a rail violation of the deactivation power sequence in response to the value of the rail voltage: at a third time, not satisfying a first condition defined by a second voltage threshold, or at a fourth time, not satisfying a second condition defined by the second voltage threshold.
15. The power supply device of claim 14, wherein the internal diagnostic circuit includes: a sequence delay check counter associated with a plurality of counter values that correspond to the first time through the fourth time.
16. The power supply device of claim 14, wherein the first condition defined by the second voltage threshold includes the value of the rail voltage being equal to or greater than the second voltage threshold, and wherein the second condition defined by the second voltage threshold includes the rail voltage being less than the second voltage threshold.
17. A method of monitoring a power sequence of a power supply system, the method comprising: receiving programming data including a first counter value and a second counter value; storing the first and second counter values in a memory device; in response to activation of an enable signal to activate a power supply system according to an activation power sequence, activating a power supply device to generate a rail voltage; detecting a rail violation of the activation power sequence in response to a value of the rail voltage: at a first time represented by the first counter value, not satisfying a first condition defined by a voltage threshold, or at a second time represented by the second counter value, not satisfying a second condition defined by the voltage threshold; and in response to the rail violation being detected, activating an interrupt signal.
18. The method of claim 17, wherein the first condition defined by the voltage threshold includes the value of the rail voltage being less than the voltage threshold, and wherein the second condition defined by the voltage threshold includes the rail voltage being greater than or equal to the voltage threshold.
19. The method of claim 17, wherein the voltage threshold is a first voltage threshold, the method further comprising: detecting deactivation of the enable signal to deactivate the power supply system according to a deactivation power sequence; and detecting a rail violation of the

deactivation power sequence in response to the value of the rail voltage: at a third time, not satisfying a first condition defined by a second voltage threshold, or at a fourth time, not satisfying a second condition defined by the second voltage threshold.

20. The method of claim 17, further comprising: detecting, by a sequence delay check counter, activation of the enable signal; triggering, by the sequence delay check counter, a rail check in response to a period of time satisfying a counter value that represents the first time; and in response to the rail check, detecting whether the value of the rail voltage satisfies the first condition defined by the voltage threshold.